INFOSYS SE REASONING VAULT • DATA SUFFICIENCY & INEQUALITIES

Mathematical Inequalities, Condition Sufficiency, Ranking & Order • Total Solved Questions: 91

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q1. How many students are in the class? Statements: I. Ramesh is ranked 12th from the top. II. Suresh is ranked 8th from the bottom. III. There are 5 students between Ramesh and Suresh.

(A) Only I and II together are sufficient
(B) Only II and III together are sufficient
(C) Only I and III together are sufficient
(D) All I, II, and III together are sufficient
(E) Even all statements together are not sufficient

Correct Answer: Option (D) : All I, II, and III together are sufficient

■ **Step-by-Step Logic:** From Statement I, Ramesh is 12th from the top. From Statement III, there are 5 students between Ramesh and Suresh, meaning Suresh's rank from the top is 18th. From Statement II, Suresh is 8th from the bottom. Since Suresh is 18th from the top and 8th from the bottom, the total number of students = 18 + 8 - 1 = 25. All three statements together are required to determine the total number of students.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q2. Question: Who among P, Q, T, V and M is exactly in the middle when they are arranged in ascending order of their heights ? Statements: I. V is taller than Q but shorter than M. II. T is taller than Q and M but shorter than P.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (E) : Both I and II are sufficient

■ **Step-by-Step Logic:** From I, we have: M > V > Q. From II, we have: T > Q, T > M, P > T. Combining the above two, we have: P>T>M>V>Q i.e. Q Clearly, M is in the middle.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q3. Question: How is 'DATE' written in the code language ? Statements: I. DEAR is written as $#@? in that code. II. TREAT is written as %?#@% in that code. III. TEAR is written as %#@? in that code.

(A) Only I and II together are sufficient
(B) Only II and III together are sufficient
(C) Only I and III together are sufficient
(D) All I, II, and III together are sufficient
(E) Even all statements together are not sufficient

Correct Answer: Option (D) : All I, II, and III together are sufficient

■ **Step-by-Step Logic:** Observing I, II and III, we find that similar letters have similar code symbols at the corresponding places in the code. So, this is direct-coding. Thus, to find the code for DATE, we need the code for D which can be obtained from I only (i.e. $.) and the codes for A, T and E which can be obtained either from II or III (@, # and % respectively).

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q4. What is the total salary of A and B? Statements: I. A's salary is ■10,000 more than B's salary, and B's salary is ■40,000. II. The difference between A's and B's salary is ■10,000.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (A) : I alone is sufficient while II alone is not sufficient

■ **Step-by-Step Logic:** Statement 1: A's salary = 40,000 + 10,000 = ■50,000 Total Salary= 50,000 + 40,000 = ■90,000 So, total salary is determined from statement I only.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q5. How many pencils does the shopkeeper sells on Sunday ? I. On Sunday he sold 12 more pencils than he sold the previous day II. He sold 28 pencils each on Thursday and Saturday

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (E) : Both I and II are sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (E) Both I and II are sufficient. All condition checks confirm consistency.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q6. What is the position of A from the left end of row? I. In a row having 15 people, A is sitting second to the right of B who is sitting at twelfth position from the right end of row. II. In a row of 20 people, C will take the eighth position from the right end of row if C and A interchange their respective positions.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (C) : Either I or II is sufficient

■ **Step-by-Step Logic:** From Statement I, B is 12 th from the right end, so A is 10th from right end so (15-10+1) 6 th from left end. From statement II, C is 8 th from right end means before changing positions, A was at 8th position from right, so (20-8+1) = 13 th from left end.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q7. What is the total salary of A and B together? Statements: I. A's salary is 20% more than B's salary. II. B's salary is ■40,000. III. A's salary is ■48,000.

(A) Only I and II together are sufficient
(B) Only II and III together are sufficient
(C) Only I and III together are sufficient
(D) All I, II, and III together are sufficient
(E) Even all statements together are not sufficient

Correct Answer: Option (B) : Only II and III together are sufficient

■ **Step-by-Step Logic:** Statement II: B's salary = ■40,000. Statement III: A's salary = ■48,000. Total salary = ■40,000 + ■48,000 = ■88,000. Statement I is unnecessary since we already have exact values.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q8. What is the total monthly salary of Ramesh? Statements: I. Ramesh's basic salary is ■40,000, and his total salary includes HRA and allowances. II. HRA is 30% of the basic salary, and allowances are ■12,000.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (E) : Both I and II are sufficient

■ **Step-by-Step Logic:** From statement I, we know the basic salary is ■40,000. From statement II, we know HRA = 30% of ■40,000 = ■12,000 and allowances = ■12,000. Thus, total salary = ■40,000 + ■12,000 + ■12,000 = ■64,000. So, both statements together are sufficient.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q9. In which year was Sanjay born? Statements: I. Sanjay is six years older than Gopal. II. Gopal's brother was born in 1982. III. Sanjay's brother is two years younger than Gopal's brother who was eight years younger than Gopal.

(A) Only I and II together are sufficient
(B) Only II and III together are sufficient
(C) Only I and III together are sufficient
(D) All I, II, and III together are sufficient
(E) Even all statements together are not sufficient

Correct Answer: Option (D) : All I, II, and III together are sufficient

■ **Step-by-Step Logic:** From Statement II, Gopal's brother was born in 1982. From Statement III, Sanjay's brother is two years younger than Gopal's brother → Born in 1984. Gopal's brother was 8 years younger than Gopal, so Gopal was born in 1974. From Statement I, Sanjay is 6 years older than Gopal, so he was born in 1968. All three statements together are required to determine the birth year of Sanjay.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q10. What is the present age of Sunita? Statements: I. Five years ago, Sunita's age was half of what it will be in 10 years. II. The difference between Sunita's present age and twice her age five years ago is 10 years.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (C) : Either I or II is sufficient

■ **Step-by-Step Logic:** From statement I: Let Sunita's present age be x. Five years ago, her age was (x - 5). In 10 years, her age will be (x + 10). Given: (x - 5) = ½ (x + 10) Solving for x, we get x = 20. From statement II: Let Sunita's age be x. Twice her age five years ago = 2(x - 5). Given: x - 2(x - 5) = 10 Solving for x, we get x = 20.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q11. Question: How many children does M have ? Statements: I. H is the only daughter of X who is wife of M. II. K and J are brothers of M.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (D) : Neither I nor II is sufficient

■ **Step-by-Step Logic:** From I, we conclude that H is the only daughter of M. But this does not indicate that M has no son. The information given in II is immaterial.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q12. 3. Is number X an even number? Statements: I. X is divisible by 4. II. X is a multiple of 6.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (C) : Either I or II is sufficient

■ **Step-by-Step Logic:** From statement I: If ■ is divisible by 4, then ■ is definitely even. So, statement I alone is sufficient. From statement II: If ■ is a multiple of 6, ■ could be 12, 18, 24, 30, ... (all even).

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q13. Can a water tank of 2500 litres capacity be filled by two inlet pipes A and B together in less than 8 hours? I. If pipe A is opened alone then it takes 12 hours to fill the tank. II. Pipe B can fill 1 litres of water per minute.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (E) : Both I and II are sufficient

■ **Step-by-Step Logic:** From the statement I, the efficiency of pipe A per hour = 2500/12 = 625/3 litres per hour From the statement II, the efficiency of pipe B = 1 × 60 = 60 litres per hour Pipe A and B together will fill in 1 hour = 625/3 + 60 = 805/3 litres per hour The time it will take to fill 2500 litres of water = more than 8 hours, therefore answer will be no Therefore, the data in both the statements I and II together is necessary to answer the question

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q14. What is the speed of a train? Statements: I. The train crosses a pole in 8 seconds. II. The train is 200 meters long.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (E) : Both I and II are sufficient

■ **Step-by-Step Logic:** Speed = Distance / Time. From statement I, we know time = 8 seconds, but we don't know the distance. From statement II, we know the length of the train is 200 meters. Using both together: Speed = 200 / 8 = 25 m/s. So, both statements together are sufficient.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q15. What is the code for "WORLD" in a certain coding language? Statements: I. "HELLO" is coded as "IFMMP". II. "NIGHT" is coded as "OJHIU". III. The coding follows a pattern where each letter is replaced by the next letter in the English alphabet.

(A) Only I and II together are sufficient
(B) Only II and III together are sufficient
(C) Only I and III together are sufficient
(D) All I, II, and III together are sufficient
(E) Even all statements together are not sufficient

Correct Answer: Option (C) : Only I and III together are sufficient

■ **Step-by-Step Logic:** Statement I: Shows how "HELLO" is coded as "IFMMP". Statement II: Shows how "NIGHT" is coded as "OJHIU". Statement III: Confirms the pattern → Each letter is replaced by the next letter. Applying this pattern to WORLD: W → X O → P R → S L → M D → E So, "WORLD" = "XPSME". Statements I and III together are sufficient.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: Data Sufficiency

Q16. What is the sum of the angles of a triangle? I. One of the angles of the triangle is 60 degrees. II. The triangle is an equilateral triangle.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (B) : II alone is sufficient while I alone is not sufficient

■ **Step-by-Step Logic:** Step 1: Checking Statement I It only tells us that one of the angles is 60 degrees, but it does not give information about the sum of all angles. However, the sum of the angles in any triangle is always 180 degrees, which is a known fact. Statement I alone is NOT sufficient, as it does not help determine whether the general rule of 180° holds. Step 2: Checking Statement II An equilateral triangle has all three angles equal to 60 degrees. Since we know it's a triangle, its angles must sum to 180 degrees by definition. Statement II alone is sufficient, as it confirms the sum of angles.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: *Data Sufficiency*

Q17. What is the value of X? Statements: I. ■ is an even number. II. ■ is a multiple of 4 and lies between 10 and 20.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (D) : Neither I nor II is sufficient

■ **Step-by-Step Logic:** Statement 1: X is even → Possible values: any even number (not sufficient). Statement 2: X is a multiple of 4 and lies between 10 and 20 → Possible values: 12, 16 (not unique). Even together, we can't determine a unique value.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: *Data Sufficiency*

Q18. On which day of the week did Raman leave for Dubai ? I. Raman didn't leave for Dubai during the weekend. II. Raman's brother left for Dubai on Friday 2 days after Raman left for Dubai

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (B) : II alone is sufficient while I alone is not sufficient

■ **Step-by-Step Logic:** From Statement II: Raman's brother left on Friday, 2 days after Raman → So Raman left on Wednesday. Statement I only tells us Raman didn't leave on the weekend but doesn't determine the exact day. Thus, Statement II alone is sufficient.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: *Data Sufficiency*

Q19. Five persons - A, B, C, D and E are sitting in a row. Who is sitting in the middle? Statements: I. B is between E and C. II. B is to the right of E. III. D is between A and E.

(A) Only I and II together are sufficient
(B) Only II and III together are sufficient
(C) Only I and III together are sufficient
(D) All I, II, and III together are sufficient
(E) Even all statements together are not sufficient

Correct Answer: Option (D) : All I, II, and III together are sufficient

■ **Step-by-Step Logic:** From I, the order is : E, B, C or C, B, E. From II, the order is : E, B. From III, the order is : A, D, E. Combining the above three, we get the order as : A, D, E, B, C. Clearly, E is sitting in the middle.

[OnlineStudy4U Topicwise Series (Data Sufficiency)]

Topic: *Data Sufficiency*

Q20. What is the speed of a train? Statements: I. The train crosses a 240-meter platform in 12 seconds. II. The length of the train is 120 meters.

(A) I alone is sufficient while II alone is not sufficient
(B) II alone is sufficient while I alone is not sufficient
(C) Either I or II is sufficient
(D) Neither I nor II is sufficient
(E) Both I and II are sufficient

Correct Answer: Option (E) : Both I and II are sufficient

■ **Step-by-Step Logic:** Statement 1: Distance covered = Train length + Platform length = 120 + 240 = 360 meters Speed = 360/12 = 30 m/s So, speed is determined. Statement 2: Only gives train length, which is not enough to calculate speed.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q21. Statements: D > C, A > B > C, D > E > F, Conclusions: I. E > C II. F > B

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (D) : Neither conclusion I nor II is valid.

■ **Step-by-Step Logic:** A > B > C < D > E > F I. Not follow II. Not follow

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q22. Statements: L≤W≤M = J, W = K Conclusions: I. M>K II. L = K

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (D) : Neither conclusion I nor II is valid.

■ **Step-by-Step Logic:** Conclusion I: M>K (False) → Given W=K and W≤M, we get K≤M, but M could be equal to K, not necessarily greater. Conclusion II: L=K (False) → Given L≤W and W=K, we get L≤K, but L could be less than K, not necessarily equal.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q23. What should be placed in place of question mark in equation B ≤ A ?N ?K ?S to make A > K definitely true.

(A) =, >, ≥
(B) =, ≥,>
(C) <, <,<
(D) <, <,=
(E) None

Correct Answer: Option (A) : =, >, ≥

■ **Step-by-Step Logic:** B≤A=N>K≥S Since A=N and N>K, we get A>K, which is definitely true. Thus, option (a) is correct.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q24. Statement: 3 ≥ 9 < 7 ≤ 10 = 2 ≤ 6 Conclusions: I. 6 > 9 II. 9 ≤ 2

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (A) : Only conclusion I is valid.

■ **Step-by-Step Logic:** Checking conclusion I: 6 > 9 From the given statement, we get: While moving from 6 towards 9, the common sign of inequalities is '>' and the given conclusions is also '6 > 9'. Clearly, C1 follows. Checking conclusion II: 9 ≤ 2 In the statement 9 < 7 ≤ 10 = 2, the common sign of inequalities between 9 and 2 is '<' whereas the given conclusion is '9 ≤ 2'. Therefore, C2 doesn't follow.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q25. A @ B means 'A is not less than B' A $ B means 'A is not more than B' A # B means 'A is neither less nor more than B' A * B means 'A is neither more than nor equal to B' A % B means 'A is neither less than nor equal to B' Statements : R#M$X*T%W@B Conclusions : I.M*W II. T@M

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (D) : Neither conclusion I nor II is valid.

■ **Step-by-Step Logic:** For conclusion I: We can clearly see the opposite sign persisting between M and W , thus no relationship between them can be established. Hence conclusion I does not follow. For conclusion II: The common sign between M and T is '<', thus M < T or T > M is the actual relationship. Hence conclusion II does not follow

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q26. Statements: M>N, N≥O, O>P Conclusions: I. M>O II. N>P

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I or II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (E) : Both the conclusion I and II are valid.

■ **Step-by-Step Logic:** Since M>N and N≥O, we can conclude M>O, so Conclusion I is true. Also, since N≥O and O>P, we get N>P, so Conclusion II is true.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q27. Statements: Y ≥ W = N; N ≥ M Conclusions: I. Y > M II. Y = M

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I nor II is valid.
(D) Neither conclusion I or II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (C) : Either conclusion I nor II is valid.

■ **Step-by-Step Logic:** Y ≥ W = N; N ≥ M I. Y > M II. Y = M So either conclusion I or II is follow

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q28. Some friends were discussing about their savings. M says that his savings are not more than that of W. W says that his savings are neither less than nor equal to S. S says that his savings are not less than O, who is not the least saver. T says that his savings are less than M but not less than P. P says that his savings are less than M. Who among the following is the highest saver?

(A) W
(B) S
(C) O
(D) P
(E) None

Correct Answer: Option (A) : W

■ **Step-by-Step Logic:** The following equation can be drawn from the given hints. P ≤ T < M ≤ W > S ≥ O

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q29. Some friends were discussing about their savings. M says that his savings are not more than that of W. W says that his savings are neither less than nor equal to S. S says that his savings are not less than O, who is not the least saver. T says that his savings are less than M but not less than P. P says that his savings are less than M. P's savings are less than how many people?

(A) 1
(B) 2
(C) 3
(D) 4
(E) More than 4

Correct Answer: Option (E) : More than 4

■ **Step-by-Step Logic:** The following equation can be drawn from the given hints. P ≤ T < M ≤ W > S ≥ O

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q30. Statement: M ≥ P < H, V > T = M Conclusions: I. V > P II. T ≥ H

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (A) : Only conclusion I is valid.

■ **Step-by-Step Logic:** M ≥ P < H ..... (i) V > T = M ..... (ii) Combining all statements, we get V > T = M ≥ P < H Thus, V > P is true. Again, T ≥ H is not true.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q31. Statements: F ≤ L > T, N = O ≥ P, F > Q > S Conclusions: I. L > T II. F > S

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (E) : Both the conclusion I and II are valid.

■ **Step-by-Step Logic:** Conclusion I: L>T (True) → Given L>T, this is directly true. Conclusion II: F>S (True) → Given F>Q>S, we get F>S.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q32. Statements: A>B, B≥C, C>D, D<E Conclusions: I. A>D II. C>E

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (A) : Only conclusion I is valid.

■ **Step-by-Step Logic:** From A>B and B≥C, we get A>C. From C>D, we get A>D, so Conclusion I is true. However, we cannot compare C and E directly, so Conclusion II is false.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q33. Some friends were discussing about their savings. M says that his savings are not more than that of W. W says that his savings are neither less than nor equal to S. S says that his savings are not less than O, who is not the least saver. T says that his savings are less than M but not less than P. P says that his savings are less than M. Which of the following statements is correct? I. O's savings are not more than W. II. M's savings are not more than W. III. P's savings are not more than W.

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Only conclusion III is valid.
(D) Neither conclusion I nor II is valid.
(E) All conclusions are valid.

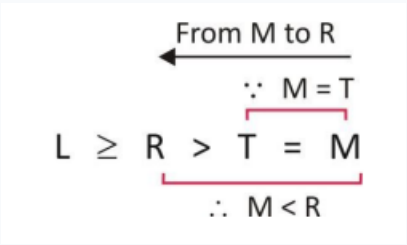
Correct Answer: Option (B) : Only conclusion II is valid.

■ **Step-by-Step Logic:** The following equation can be drawn from the given hints. P ≤ T < M ≤ W > S ≥ O

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q34. 'A © B' means 'A is either smaller than or equal to B.' 'A % B' means 'A is neither smaller than nor equal to B.' 'A @ B' means 'A is either greater than or equal to B.' 'A $ B' means 'A is neither greater than nor equal to B.' 'A # B' means 'A is neither greater than nor smaller than B.' Statements: L @ R, R % T, T # M Conclusions: I. M $ R II. T $ L



(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (E) : Both the conclusion I and II are valid.

■ **Step-by-Step Logic:** Conclusion I : [IMAGE: images/img_b700b05a9b66.png] Conclusion II : [IMAGE: images/img_6d785d48d614.png] Hence, both Conclusions I and II are true.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q35. Statements: M = X < Z ≥ W = N ≤ Q < T ≤ V = U Conclusions: I. V ≥ W II. T ■ U

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (B) : Only conclusion II is valid.

■ **Step-by-Step Logic:** Conclusion I: V ≥ W → True because W≤Q<T≤V, so V≥W. Conclusion II: T ■ U → True because T≤V and V=U, so T≤U, meaning T is not greater than U.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q36. Statement: P >S, P < Q, R ≤ Q, R < T Conclusions: I. Q > S II. Q = T

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

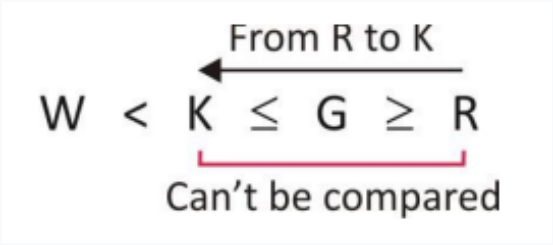
Correct Answer: Option (A) : Only conclusion I is valid.

■ **Step-by-Step Logic:** Conclusion I: Q>S Given P>S and P<Q, we can infer: Q>P>S So, Q>S is true Conclusion II: Q=T Given R≤Q and R<T, we can infer: Q≥R<T This does not guarantee Q=T, so Conclusion II is false

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q37. 'A © B' means 'A is either smaller than or equal to B.' 'A % B' means 'A is neither smaller than nor equal to B.' 'A @ B' means 'A is either greater than or equal to B.' 'A $ B' means 'A is neither greater than nor equal to B.' 'A # B' means 'A is neither greater than nor smaller than B.' Statements: W $ K, K © G, G @ R Conclusions: I. R $ K II. G % W



(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (B) : Only conclusion II is valid.

■ **Step-by-Step Logic:** Conclusion I: [IMAGE: images/img_1d1d94b6778c.png] Conclusion II: [IMAGE: images/img_ec2c6cc55e8e.png] Hence, only Conclusion II is true.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q38. Statement: P ≤ Q < S = T ≥ U ≥ W < Z Conclusion: I. S > W, II. W = T

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (C) : Either conclusion I or II is valid.

■ **Step-by-Step Logic:** For conclusion I and II: S > W and W = T From the given statement, we get: S = T ≥ U ≥ W Here, the common sign between S and W is '≥' and the given conclusions are S > W and W = S. Moreover, we are aware that 'S = T' which means we can replace T with S in conclusion 2. Hence, either conclusion I or conclusion II follows.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: *Inequalities*

Q39. L ≤ O > V = E ≥ S Which of the following ones is correct? 1) L ≤ V 2) O = E 3) O > S 4) S ≥ L

(A) Only 1
(B) Only 2
(C) Only 3
(D) Only 3 and 4
(E) None

Correct Answer: Option (C) : Only 3

■ **Step-by-Step Logic:** L ≤ O > V = E ≥ S We can compare O and S. which shows that the option 3rd is correct because the common symbol between O and S is '>'.

[OnlineStudy4U Topicwise Series (Inequalities)]

Topic: Inequalities

Q40. A @ B means 'A is not less than B' A $ B means 'A is not more than B' A # B means 'A is neither less nor more than B' A * B means 'A is neither more than nor equal to B' A % B means 'A is neither less than nor equal to B' Statements : V@I#E*D ; N$E%B#F Conclusions : I. V%F II. V@N

(A) Only conclusion I is valid.
(B) Only conclusion II is valid.
(C) Either conclusion I or II is valid.
(D) Neither conclusion I nor II is valid.
(E) Both the conclusion I and II are valid.

Correct Answer: Option (E) : Both the conclusion I and II are valid.

■ **Step-by-Step Logic:** Conclusions: I. V > F II. V ≥ N For conclusion I: From both the statements we get: V ≥ I = E > B = F The common sign between V and F is '>', thus V > F is true. Hence conclusion I follows. For conclusion II: From both the statements we get: V ≥ I = E ≥ N The common sign between V and N is '≥', thus V ≥ N is the actual relationship. Hence conclusion II follows.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q41. In a line of people, Rohit is ranked 12th from the top, and Ankit is ranked 20th from the bottom. If there are 30 people in the line, how many people are between Rohit and Ankit?

(A) 5
(B) 7
(C) 0
(D) 9

Correct Answer: Option (C) : 0

■ **Step-by-Step Logic:** Rohit position from bottom = (30-12)+1 = 19 Ankit position from bottom = 20 Number of people between Rohit and Ankit = 0

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q42. Arjun is older than Sumit but younger than Rahul. Vikas is younger than Rahul but older than Arjun. Ramesh is the oldest among them. Who is the third oldest?

(A) Sumit
(B) Arjun
(C) Rahul
(D) Vikas

Correct Answer: Option (D) : Vikas

■ **Step-by-Step Logic:** From the given conditions: Arjun > Sumit, but Arjun < Rahul. → Rahul > Arjun > Sumit. Vikas is younger than Rahul but older than Arjun. → Rahul > Vikas > Arjun > Sumit. Ramesh is the oldest. Final order: Ramesh > Rahul > Vikas > Arjun > Sumit. The third oldest is Vikas.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q43. Six people—P, Q, R, S, T, and U—are standing in a line from left to right. P is taller than S but shorter than Q. R is taller than Q but shorter than U. T is the tallest. Who is the fourth tallest?

(A) P
(B) Q
(C) R
(D) S

Correct Answer: Option (B) : Q

■ **Step-by-Step Logic:** From the given conditions: P > S, but P < Q → Q > P > S. R > Q, but R < U → U > R > Q. T is the tallest. Final order: T > U > R > Q > P > S. The fourth tallest person is Q.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q44. Shreya is taller than Meena but shorter than Ritu. Kajal is shorter than Ritu but taller than Meena. Payal is the shortest among all. Who is the third tallest?

(A) Shreya
(B) Ritu
(C) Meena
(D) Kajal

Correct Answer: Option (D) : Kajal

■ **Step-by-Step Logic:** From the given conditions: Shreya > Meena but Shreya < Ritu. → Ritu > Shreya > Meena. Kajal is shorter than Ritu but taller than Meena. → Ritu > Kajal > Meena. Payal is the shortest. Arranging all: Ritu > Shreya > Kajal > Meena > Payal. The third tallest person is Kajal.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q45. In a row of students, Raj is 15th from the left end and 12th from the right end. How many students are there in the row?

(A) 26
(B) 28
(C) 25
(D) 30

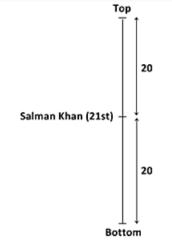
Correct Answer: Option (A) : 26

■ **Step-by-Step Logic:** Total students in the row = (Position from Left) + (Position from Right) – 1 = 15 + 12 - 1 = 26

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q46. During assembly the students are standing in a line. Salman Khan is 21st in order from both the ends. How many boys are there in the class?



(A) 30
(B) 40
(C) 31
(D) 41

Correct Answer: Option (D) : 41

■ **Step-by-Step Logic:** [IMAGE: images/img_1b724a1ed51a.png] Therefore, total number of students = 20 + Salman Khan + 20 = 41

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q47. In a row of 45 students, Aditi is ranked 18th from the left. Priya is 24th from the right. If there are 5 students between Aditi and Priya, how many students are there in the row?

(A) 45
(B) 47
(C) 50
(D) 53

Correct Answer: Option (B) : 47

■ **Step-by-Step Logic:** Total students in the row = (Aditi's rank from the left) + (Priya's rank from the right) + (Students between them) = 18 + 24 + 5 = 47

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q48. Rahul is heavier than Suresh but lighter than Mohan. Kunal is lighter than Rahul but heavier than Suresh. Deepak is heavier than all except Mohan. Who is the second lightest?

(A) Rahul
(B) Kunal
(C) Suresh
(D) Mohan

Correct Answer: Option (B) : Kunal

■ **Step-by-Step Logic:** Rahul is heavier than Suresh but lighter than Mohan. → Mohan > Rahul > Suresh Kunal is lighter than Rahul but heavier than Suresh. → Mohan > Rahul > Kunal > Suresh Deepak is heavier than all except Mohan. → Mohan > Deepak > Rahul > Kunal > Suresh Solution: The second lightest is Kunal.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q49. In a north-facing row of 45 students, Neha is 13th from the left, and Priya is 15th from the right. There are 4 students between Priya and Rani, who is equidistant to Neha. What is Rani's position from the right?

(A) 25
(B) 20
(C) 30
(D) 32

Correct Answer: Option (B) : 20

■ **Step-by-Step Logic:** Priya’s position from the left = 45−15+1=31. Rani’s position from the left (5 places ahead of Priya) = 31−5=26. Check equidistance: ■26−13■=13 (same as Neha's position). Rani’s position from the right = 45−26+1=20.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q50. Five students—Amit, Bhavesh, Chetan, Dinesh, and Esha—took an exam. Amit scored higher than Chetan but lower than Bhavesh. Dinesh scored more than Amit but less than Esha. Esha scored the highest. Who got the second highest marks?

(A) Dinesh
(B) Chetan
(C) Amit
(D) Esha

Correct Answer: Option (A) : Dinesh

■ **Step-by-Step Logic:** From the given conditions: Amit > Chetan, but Amit < Bhavesh. → Bhavesh > Amit > Chetan. Dinesh > Amit but Dinesh < Esha. → Esha > Dinesh > Amit. Esha scored the highest. Final order: Esha > Dinesh > Bhavesh > Amit > Chetan. The second highest scorer is Dinesh.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q51. In a class of 50 students, Priya ranks 14th from the top. What is her rank from the bottom?

(A) 35
(B) 37
(C) 40
(D) 45

Correct Answer: Option (B) : 37

■ **Step-by-Step Logic:** Rank from the bottom = Total students - Rank from the top + 1 = 50 - 14 + 1 = 37

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q52. If the rank of Rohan is 7th from the top in a group of 50 candidates and 2 new candidates are added just before and after him, what will be his new rank from bottom?

(A) 44
(B) 45
(C) 46
(D) 48

Correct Answer: Option (C) : 46

■ **Step-by-Step Logic:** Adding of candidates just before Rohan will not change his rank from bottom. Before addition, the rank of Rohan from bottom is = (50 - 7) + 1 = 44 Now adding 2 candidates just after Rohan will increase his rank by 2 positions. Hence; the new rank of Rohan from bottom will be = 44 + 2 = 46

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q53. Neha is heavier than Rina but lighter than Simran. Aisha is lighter than Neha but heavier than Rina. Jaya is heavier than all except Simran. Who is the third heaviest?

(A) Rina
(B) Simran
(C) Neha
(D) Jaya

Correct Answer: Option (C) : Neha

■ **Step-by-Step Logic:** From the given conditions: Neha > Rina, but Neha < Simran → Simran > Neha > Rina. Aisha is lighter than Neha but heavier than Rina → Simran > Neha > Aisha > Rina. Jaya is heavier than all except Simran → Simran > Jaya > Neha > Aisha > Rina. The third heaviest person is Neha.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q54. Six persons – Sumer, Anant, Parth, Rajat, Karan and Gaurav, played in a cricket match and scored different number of runs. Sumer scored more runs than only Parth and Gaurav. Anant scored less runs than Karan, who was not highest scorer. The Second highest person scored 78 runs. How many person(s) scored more runs than Parth?

Score	Person	Runs
Highest	Rajat	
↓	Karan	78
	Anant	
	Sumer	
	Parth/Gaurav	
Lowest	Gaurav/Parth	

(A) 1
(B) 2
(C) 3
(D) More than 3

Correct Answer: Option (D) : More than 3

■ **Step-by-Step Logic:** [IMAGE: images/img_77f39ef55721.png] Following the final solution we can say that either 4 or 5 persons scored more runs than Parth.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q55. In a line of 60 people, Sam is ranked 21st from the top. How many people are ahead of him and how many are behind him?

(A) 20, 39
(B) 20,40
(C) 22,39
(D) 22,40

Correct Answer: Option (A) : 20, 39

■ **Step-by-Step Logic:** People ahead of Sam = Sam's rank - 1 = 21 - 1 = 20 People behind Sam = Total - Sam's rank = 60 - 21 = 39

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q56. In a nursery, I saw a group of small rose trees arranged in a row. However, one rose tree caught my attention very much. I started counting its rank from the left hand side and found out to be 12. When I started counting from right hand side, it was again 12!! What is the total number of rose trees that were arranged in that row?

(A) 21
(B) 25
(C) 32
(D) 23

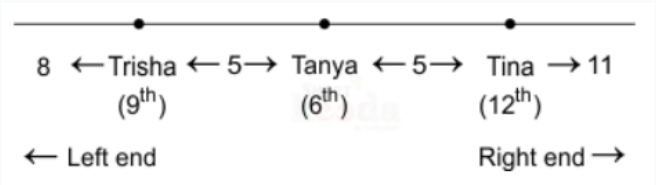
Correct Answer: Option (D) : 23

■ **Step-by-Step Logic:** As the rank of the tree is same from either side, the total number of trees will be 12 + 12 – 1 = 23

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q57. In a north facing row of NCC Cadets, Trisha is 9th from the left end and Tina is 12th from the right end. There are 5 cadets between Trisha and Tanya who is equidstant to Tina. Find how many cadets are there in the row?



(A) 34
(B) 31
(C) 32
(D) 33

Correct Answer: Option (C) : 32

■ **Step-by-Step Logic:** [IMAGE: images/img_3482b1a61c42.png] Adding all the persons in the above image, we get: 8 + 1(Trisha) + 5 + 1(Tanya) + 5 + 1(Tina) + 11 = 32 Thus there are 32 cadets in the row.

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q58. In a north-facing row of students, Ajay is 18th from the left and Vijay is 22nd from the right. If there are 10 students between them, find the total number of students in the row.

(A) 40
(B) 45
(C) 50
(D) 55

Correct Answer: Option (C) : 50

■ **Step-by-Step Logic:** Total students = Ajay's position from left + Vijay's position from right + students between them = 18 + 22 + 10 = 50

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q59. Aman is standing in a queue at the 10th position from the front. If he is 15th from the back, how many people are in the queue?

(A) 30
(B) 75
(C) 50
(D) 24

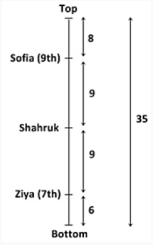
Correct Answer: Option (D) : 24

■ **Step-by-Step Logic:** Total people in the queue = (Position from Front) + (Position from Back) - 1 = 10 + 15 - 1 = 24

[OnlineStudy4U Topicwise Series (Ranking Test)]

Topic: Ranking Test

Q60. In a class of 35 students, Ziya is placed 7th from the bottom where as Sofia is placed 9th from the top. Shahruk is placed in between the two. What is Ziya's position from Shahruk?



(A) 10
(B) 15
(C) 19
(D) 21

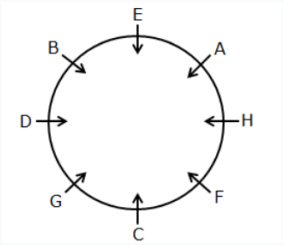
Correct Answer: Option (A) : 10

■ **Step-by-Step Logic:** As seen in the figure, Shahruk is between Sofia and Ziya. It's given that Ziya is 7th from the bottom and Sofia is 9th from the top. Therefore, number of persons between Sofia and Ziya = 35-(9+7) = 19 [IMAGE: images/img_446d663e7608.png] Sharuk's position between Sofia and Ziya = 19-1/2 = 10. Hence, Shahruk is at the middle ie. at 10th position from both. Ziya, therefore, is at the 10th position from Shahruk

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q61. Eight persons viz. A, B, C, D, E, F, G, and H are sitting around a circular table such that all of them are facing towards the center. The consecutive alphabetically named persons are not sitting adjacent to each other. Two persons sit between A and C, who faces E. Only one person sits between A and F. The number of persons sitting between E and F is twice the number of persons sitting between C and H when counted from the right of both F and C. D sits exactly between B and G, who doesn't sit adjacent to E. What is the position of F with respect to H?



(A) Fourth to the right
(B) Immediate left
(C) Second to the left
(D) Third to the right

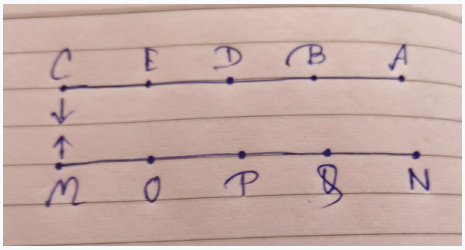
Correct Answer: Option (B) : Immediate left

■ **Step-by-Step Logic:** [IMAGE: images/img_31e2899d3527.png]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q62. Each of these questions are based on the information given below: A ,B, C, D and E are five men sitting in a line facing to south - while M, N, O, P and Q are five ladies sitting in a second line parallel to the first line and are facing to North. B who is just next to the left of D, is opposite to Q. C and N are diagonally opposite to each other. E is opposite to O who is just next right of M. P who is just to the left of Q, is opposite to D. M is at one end of the line. Which of the following pair is diagonally opposite to each other?



(A) EQ
(B) BO
(C) AN
(D) AM

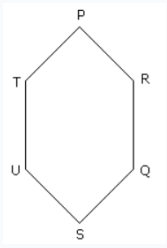
Correct Answer: Option (D) : AM

■ **Step-by-Step Logic:** [IMAGE: images/img_c72e8901af5c.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q63. Six friends P, Q, R, S, T and U are sitting around the hexagonal table each at one corner and are facing the centre of the hexagonal. P is second to the left of U. Q is neighbour of R and S. T is second to the left of S. Which of the following are the neighbours of P ?



(A) U,P
(B) U,R
(C) T,R
(D) None

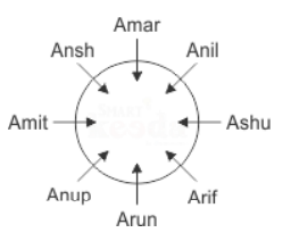
Correct Answer: Option (C) : T,R

■ **Step-by-Step Logic:** [IMAGE: images/img_cc821d83fcc7.png]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q64. Eight persons – Amar, Ansh, Arun, Amit, Ashu, Anup, Arif and Anil, are sitting around a circular table facing the centre but not necessarily in the same order. Arif is to the immediate left of Ashu. Anup is to the immediate left of Arun. Amar is the neighbour of Ansh. Amit is between Anup and Ansh. Arif is not the neighbour of Ansh or Anil. What is the position of Arif with respect to Amit?



(A) Third to the right
(B) Second to the left
(C) Fourth to the right
(D) Fourth to the left

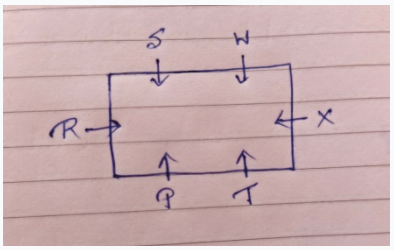
Correct Answer: Option (A) : Third to the right

■ **Step-by-Step Logic:** [IMAGE: images/img_3bd3748bb523.png] Following the final solution, we can say that Arif is third to the right of Amit.

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q65. Six persons namely P, R, S, T, W, and X are sitting around a rectangular table such that one sits at each of the shorter sides and two sit at each of the longer sides of the table facing towards the center of the table. W sits on the longer side and two persons sit between P and W. R sits opposite to X and is near P. T doesn't sit near W. X doesn't sit immediately left of T. Pick the odd one out.



(A) T
(B) P
(C) W
(D) None

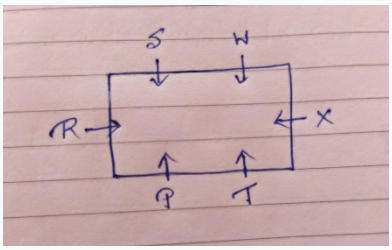
Correct Answer: Option (D) : None

■ **Step-by-Step Logic:** [IMAGE: images/img_063b4d16fe07.jpg] Clearly, all of them are facing inside and sitting on the longer sides. None is odd.

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q66. Six persons namely P, R, S, T, W, and X are sitting around a rectangular table such that one sits at each of the shorter sides and two sit at each of the longer sides of the table facing towards the center of the table. W sits on the longer side and two persons sit between P and W. R sits opposite to X and is near P. T doesn't sit near W. X doesn't sit immediately left of T. How many persons sit between P and T when counted from the left of P?



(A) 1
(B) 2
(C) 3
(D) 4

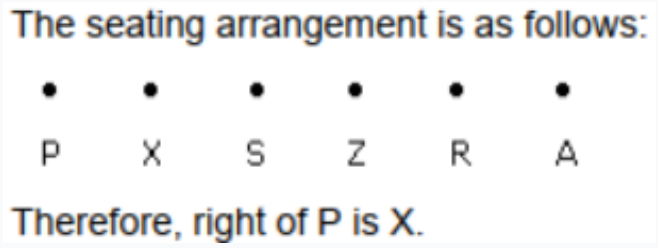
Correct Answer: Option (D) : 4

■ **Step-by-Step Logic:** [IMAGE: images/img_d3d6d1a122f3.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q67. A, P, R, X, S and Z are sitting in a row. S and Z are in the centre. A and P are at the ends. R is sitting to the left of A. Who is to the right of P ?



(A) X
(B) S
(C) Z
(D) A

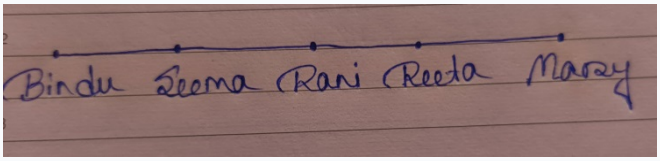
Correct Answer: Option (A) : X

■ **Step-by-Step Logic:** [IMAGE: images/img_ddae95ff9141.png]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q68. Five girls are sitting on a bench to be photographed. Seema is to the left of Rani and to the right of Bindu. Mary is to the right of Rani. Reeta is between Rani and Mary. Who is sitting immediate right to Reeta ?



(A) Bindu
(B) Rani
(C) Mary
(D) Seema

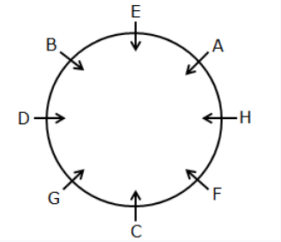
Correct Answer: Option (C) : Mary

■ **Step-by-Step Logic:** [IMAGE: images/img_f0201711f861.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q69. Eight persons viz. A, B, C, D, E, F, G, and H are sitting around a circular table such that all of them are facing towards the center. The consecutive alphabetically named persons are not sitting adjacent to each other. Two persons sit between A and C, who faces E. Only one person sits between A and F. The number of persons sitting between E and F is twice the number of persons sitting between C and H when counted from the right of both F and C. D sits exactly between B and G, who doesn't sit adjacent to E. Who among the following person sits second to the left of C?



(A) The one who sits to the immediate right of E
(B) G
(C) The one who sits third to the right of A
(D) H

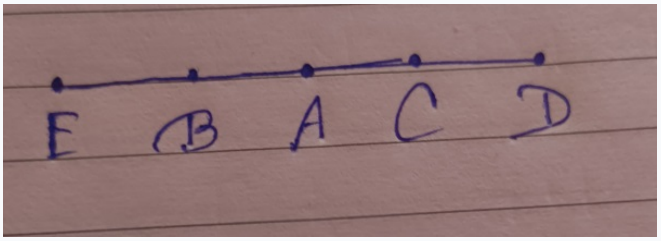
Correct Answer: Option (C) : The one who sits third to the right of A

■ **Step-by-Step Logic:** [IMAGE: images/img_75b4eb26828e.png]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q70. A, B, C, D and E are sitting on a bench. A is sitting next to B, C is sitting next to D, D is not sitting with E who is on the left end of the bench. C is on the second position from the right. A is to the right of B and E. A and C are sitting together. In which position A is sitting ?



(A) Between B and C
(B) Between B and D
(C) Between E and D
(D) Between C and D

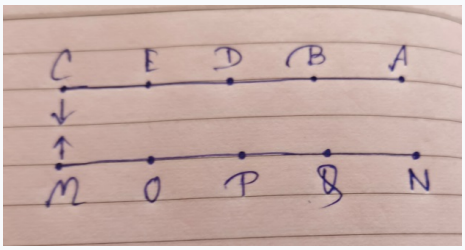
Correct Answer: Option (A) : Between B and C

■ **Step-by-Step Logic:** [IMAGE: images/img_ac3cca37717f.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q71. Each of these questions are based on the information given below: A ,B, C, D and E are five men sitting in a line facing to south - while M, N, O, P and Q are five ladies sitting in a second line parallel to the first line and are facing to North. B who is just next to the left of D, is opposite to Q. C and N are diagonally opposite to each other. E is opposite to O who is just next right of M. P who is just to the left of Q, is opposite to D. M is at one end of the line. Who is sitting just opposite to N ?



(A) B
(B) A
(C) C
(D) D

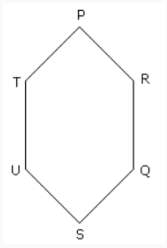
Correct Answer: Option (B) : A

■ **Step-by-Step Logic:** [IMAGE: images/img_e73946ee30b9.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q72. Six friends P, Q, R, S, T and U are sitting around the hexagonal table each at one corner and are facing the centre of the hexagonal. P is second to the left of U. Q is neighbour of R and S. T is second to the left of S. Which one is sitting opposite to P ?



(A) R
(B) S
(C) T
(D) Q

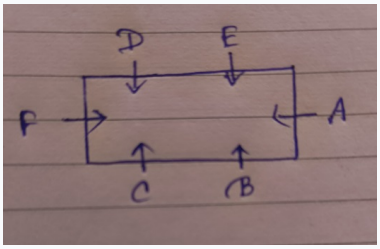
Correct Answer: Option (B) : S

■ **Step-by-Step Logic:** [IMAGE: images/img_3f7204d8221a.png]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q73. In a group discussion A, B, C, D, E and F are sitting around a rectangular table. A is sitting on a side alone. B, C are sitting on the same side. F is facing A. D is the immediate left of F. C is facing D. Who is immediate right of A?



(A) B
(B) F
(C) D
(D) E

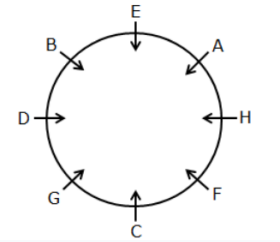
Correct Answer: Option (D) : E

■ **Step-by-Step Logic:** [IMAGE: images/img_a6561ae307e6.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q74. Eight persons viz. A, B, C, D, E, F, G, and H are sitting around a circular table such that all of them are facing towards the center. The consecutive alphabetically named persons are not sitting adjacent to each other. Two persons sit between A and C, who faces E. Only one person sits between A and F. The number of persons sitting between E and F is twice the number of persons sitting between C and H when counted from the right of both F and C. D sits exactly between B and G, who doesn't sit adjacent to E. Find out the series that follow the same logic. DG FA BC ___



(A) HD
(B) FE
(C) HB
(D) AG

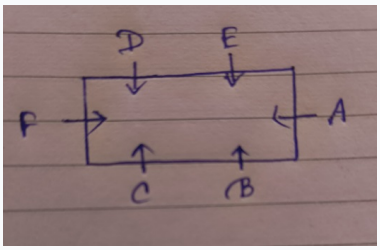
Correct Answer: Option (A) : HD

■ **Step-by-Step Logic:** [IMAGE: images/img_47587c5bb84f.png]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q75. In a group discussion A, B, C, D, E and F are sitting around a rectangular table. A is sitting on a side alone. B, C are sitting on the same side. F is facing A. D is the immediate left of F. C is facing D. Who is between B and F?



(A) C
(B) D
(C) A
(D) E

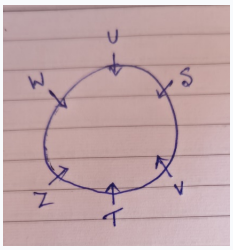
Correct Answer: Option (A) : C

■ **Step-by-Step Logic:** [IMAGE: images/img_7b41770eacff.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q76. Six people - S, T, U, V, W and Z are sitting around a circular table facing towards centre. Only one person is sitting between T and W. Z is an immediate neighbour of W. S is sitting third to left of Z. U is an immediate neighbour of W. V is not sitting immediate left of T. How many people are sitting between U and Z, when counted from the right of Z?



(A) 1
(B) 2
(C) 3
(D) 4

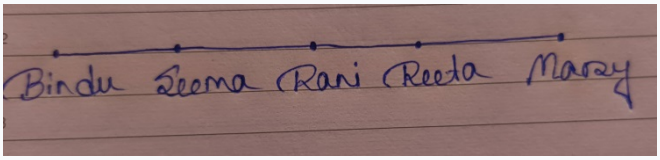
Correct Answer: Option (C) : 3

■ **Step-by-Step Logic:** [IMAGE: images/img_c8b581dc4168.jpg] T,V and S are sitting between U and Z when counted from the right of Z.

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q77. Five girls are sitting on a bench to be photographed. Seema is to the left of Rani and to the right of Bindu. Mary is to the right of Rani. Reeta is between Rani and Mary. Who is in the middle of the photograph ?



(A) Bindu
(B) Rani
(C) Seema
(D) Reeta

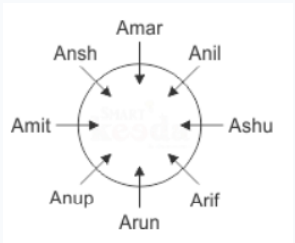
Correct Answer: Option (B) : Rani

■ **Step-by-Step Logic:** [IMAGE: images/img_e08dce26dd83.jpg]

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q78. Eight persons – Amar, Ansh, Arun, Amit, Ashu, Anup, Arif and Anil, are sitting around a circular table facing the centre but not necessarily in the same order. Arif is to the immediate left of Ashu. Anup is to the immediate left of Arun. Amar is the neighbour of Ansh. Amit is between Anup and Ansh. Arif is not the neighbour of Ansh or Anil. How many persons were sitting between Amit and Anil, when counted from the left of Anil?



(A) 1
(B) 2
(C) 3
(D) 4

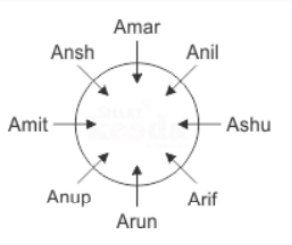
Correct Answer: Option (D) : 4

■ **Step-by-Step Logic:** [IMAGE: images/img_1f2dfe982c74.png] Following the final solution, we can say that four persons were sitting between Amit and Anil, when counted from the left of Anil.

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q79. Eight persons – Amar, Ansh, Arun, Amit, Ashu, Anup, Arif and Anil, are sitting around a circular table facing the centre but not necessarily in the same order. Arif is to the immediate left of Ashu. Anup is to the immediate left of Arun. Amar is the neighbour of Ansh. Amit is between Anup and Ansh. Arif is not the neighbour of Ansh or Anil. Which of the following person sits opposite to the one who sits third to the left of Anil?



(A) Anup
(B) Arif
(C) Amar
(D) Arun

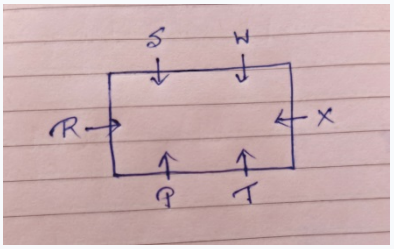
Correct Answer: Option (C) : Amar

■ **Step-by-Step Logic:** [IMAGE: images/img_e1c29cebbd5c.png] In the following common it is clear that Amar sits opposite to the one who sits third to the left of Anil.

[OnlineStudy4U Topicwise Series (Seating Arrangement)]

Topic: Seating Arrangement

Q80. Six persons namely P, R, S, T, W, and X are sitting around a rectangular table such that one sits at each of the shorter sides and two sit at each of the longer sides of the table facing towards the center of the table. W sits on the longer side and two persons sit between P and W. R sits opposite to X and is near P. T doesn't sit near W. X doesn't sit immediately left of T. Who sits second to the right of P?



(A) T
(B) R
(C) X
(D) S

Correct Answer: Option (C) : X

■ **Step-by-Step Logic:** [IMAGE: images/img_3bd5d69e8108.jpg]

[College ITM Mock Paper 1]

Topic: Reasoning Ability

Q81. Data Sufficiency. For integers x and y, is x > y? I. $x^2 > y^2$. II. x + y > 0.

(A) I alone is sufficient
(B) II alone is sufficient
(C) Both together are sufficient, neither alone
(D) Even both

Correct Answer: Option (D) : Even both

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both. All condition checks confirm consistency.

[College ITM Mock Paper 2]

Topic: Reasoning Ability

Q82. Data Sufficiency. Is the integer n divisible by 6? I. n is divisible by 4. II. n is divisible by 9.

(A) I alone is sufficient
(B) II alone is sufficient
(C) Both together are sufficient, neither alone
(D) Even both

Correct Answer: Option (D) : Even both

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both. All condition checks confirm consistency.

[College ITM Mock Paper 2]

Topic: Reasoning Ability

Q83. Data Sufficiency. What is the cost price of an article? I. It was sold at a profit of 20%. II. If sold at ■150 more, the profit would have been 32%.

(A) I alone is sufficient
(B) II alone is sufficient
(C) Both together are sufficient, neither alone
(D) Even both

Correct Answer: Option (D) : Even both

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both. All condition checks confirm consistency.

[College ITM Mock Paper 3]

Topic: Reasoning Ability

Q84. Data Sufficiency. What is the two-digit number N? I. The sum of its digits is 9. II. The number obtained by reversing its digits is 27 less than N.

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Each statement alone is sufficient

Correct Answer: Option (D) : Each statement alone is sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Each statement alone is sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 3]

Topic: Reasoning Ability

Q85. Data Sufficiency. Is the integer x even? I. x/2 is an integer. II. x^2 is even.

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Each statement alone is sufficient

Correct Answer: Option (D) : Each statement alone is sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Each statement alone is sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 4]

Topic: Reasoning Ability

Q86. Data Sufficiency. What is the value of x? I. $x^2 - 5x + 6 = 0$ II. $x > 2$

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Even both statements together are not sufficient

Correct Answer: Option (D) : Even both statements together are not sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both statements together are not sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 4]

Topic: Reasoning Ability

Q87. Data Sufficiency. In triangle ABC, what is the measure of angle B? I. Angle A = 50°, and the triangle is isosceles with AB = AC. II. Angle C = 65°.

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Even both statements together are not sufficient

Correct Answer: Option (D) : Even both statements together are not sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both statements together are not sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 5]

Topic: Reasoning Ability

Q88. Data Sufficiency. If x and y are positive integers, what is the remainder when x is divided by y? I. x = 5y + 3 II. y = 7

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Even both statements together are not sufficient

Correct Answer: Option (D) : Even both statements together are not sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both statements together are not sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 5]

Topic: Reasoning Ability

Q89. Data Sufficiency. In how many days can A and B complete a job together? I. A is twice as efficient as B. II. A alone can complete the job in 12 days.

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Even both statements together are not sufficient

Correct Answer: Option (D) : Even both statements together are not sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both statements together are not sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 6]

Topic: Reasoning Ability

Q90. Data Sufficiency. Is the product a × b positive? I. a + b > 0. II. a – b > 0.

(A) Statement I alone is sufficient, but II alone is not
(B) Statement II alone is sufficient, but I alone is not
(C) Both statements together are sufficient, but neither alone is
(D) Even both statements together are not sufficient

Correct Answer: Option (D) : Even both statements together are not sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both statements together are not sufficient. All condition checks confirm consistency.

[College ITM Mock Paper 6]

Topic: *Reasoning Ability*

Q91. Data Sufficiency. Is the positive integer n a perfect square? I. n has exactly 3 positive divisors. II. n is divisible by 9.

(A) Statement I alone is sufficient, but II alone is not

(B) Statement II alone is sufficient, but I alone is not

(C) Both statements together are sufficient, but neither alone is

(D) Even both statements together are not sufficient

Correct Answer: Option (D) : Even both statements together are not sufficient

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (D) Even both statements together are not sufficient. All condition checks confirm consistency.